Straw Burns In Fields, Forever?
Displacement Effects of a Seasonal Burning Ban in Thailand

Iddo Glass

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Introduction and contributions

Context

Data and empirical strategy

Results

Estimating Evasion

Introduction

→ Governments in low-income settings struggle to enforce environmental regulations

→ Targeted enforcement could be more cost-effective

- But could lead to displacement of activities

→ I evaluate seasonal bans on open-burning in Northern Thailand

- I find 28% reduction in fires throughout January to May
- Evidence of temporal displacement
  - Increase in fires when bans are over
  - Evasion within the day to avoid satellite detection

Contributions

→ Policies to reduce forest and crop burning [e.g., Jack et al. (2024) and Nguyen Huy, Adjognon, and Soest (2024)]

→ Specifically, burning bans [Balboni, Burgess, and Olken (Forthcoming), Cao and Ma (2023), and Dipoppa and Gulzar (2024)]

- I show that while bans may reduce fires overall, they can displace the timing of fires

→ Environmental regulations in low-income settings [e.g., Ashraf, Glaeser, and Ponzetto (2016) and Greenstone and Hanna (2014)]

→ The use of remote sensing in those contexts [e.g., Assunção, Gandour, and Rocha (2023) and Saavedra (2025)]

- I show that the impact of such systems can be undermined by behavioural responses

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Haze season in Southeast Asia

→ Extreme levels of air pollution during the dry season cause massive health damages

→ Driven by meteorological conditions and an increase in biomass burning

- Forest burning is often done to collect forest products (e.g., "earthstar" mushrooms)
- Crop burning is the cheapest and easiest way to dispose crop residue

▸ Fires by month ▸ Pollution in Chiang Mai

Northern Thailand Burning Ban

→ Starting from 2013, nine provinces in Northern Thailand banned open-burning for around two months every year

→ The provincial authorities chose the dates, usually between February and April

→ Enforcement became stricter from 2017, but remained imperfect [Nikam, Archer, and Nopsert (2021) and Pirard and Charoenpanwutikul (2023)]

→ Monitoring was partly done using NASA's FIRMS (based on satellites) [Smith (2024)]

▸ Dates of bans

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Data

→ Admin units - provinces in Thailand and Laos, districts in Myanmar

→ Fires from FIRMS

- Main outcome - fires per 100 km^2, normalised by the unit's area
- Focusing on MODIS - 1 km resolution, Aqua and Terra satellites
- VIIRS (375m) lacks pre-treatment

→ Climatic variables from ERA5

→ Elevation, land use, distances to roads and urban centres, crop production

Descriptive statistics | Area of interest | Northern Thailand provinces | Fires by provinces

Empirical strategy (1)

Synthetic control method

Estimating the causal effect of the ban-centred policy:

$$\tau_{i,t} = Y^1_{i,t} - Y^0_{i,t}$$

→ Synthetic control method to generate a counterfactual for each treated unit i

→ Weighted combinations of the donor units, that resembles each i before the treatment

→ Add a term to penalize units that are dissimilar to each i [Abadie and L'Hour (2021)]

▸ Optimization problem ▸ Obtaining ATE

Empirical strategy (2)

Inference

→ I generate a distribution of "placebo" effects for donor units [Abadie (2021)]

→ Examining $\hat{T}$ - the ratio of mean squared prediction errors after to before the treatment

→ H_0 - $\hat{T}$ for the treated units is not unusually large compared to the placebos

- p-value - the share of $\hat{T}_{placebo}$ that are larger or equal to $\hat{T}_{treated}$

▸ Test statistic

Empirical strategy (3)

Application

→ Generate weights using pre-treatment period (2004-2012)

→ Evaluate each enforcement period separately - "weak" (2013-16) and "strict" (2017-20)

- Compared to the pre-treatment period

→ Main specification examines aggregate effects from January to May

→ For each specification, each unit has a "new" synthetic control

- To examine displacement - month-specific SC
- Heterogeneity - outcome-specific SC
- Effects by treatment status the ban - examine "half-months"

▸ Evaluating the effect by treatment status

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Review of results

→ From January to May, fires in Northern Thailand reduce by 28% when the ban is enforced

- Null effect in the weak enforcement period

→ Burning season is extended

→ When examining by treatment status - increase after the ban and no effect during it

- Concerns about data accuracy and pooling of effects

→ Results are robust to various specifications

▸ Robustness tests ▸ Effects by treatment status ▸ Weights

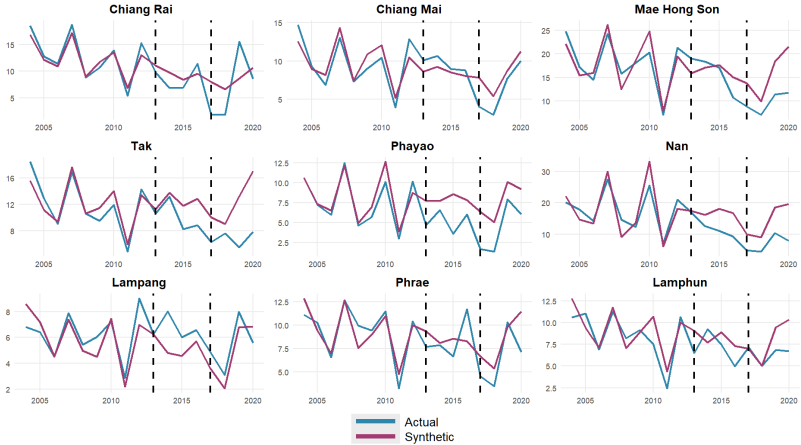
Main specification

Effect on total number of fires per 100 km^2

	Strict enforcement (2017–2020) (1)	Weak enforcement (2013–2016) (2)
Jan–May	-3.24^{*} [-28.0%]	-1.06 [-7.2%]
	(0.08)	(0.30)
January	-0.07 [37.4%]	-0.06 [25.9%]
	(0.45)	(0.53)
February	-0.62 [-17.4%]	-0.24 [-13.0%]
	(0.19)	(0.44)
March	-2.42^{*} [-20.6%]	-0.50 [5.9%]
	(0.08)	(0.21)
April	0.18 [88.0%]	0.56 [79.3%]
	(0.14)	(0.25)
May	0.29^{**} [373.8%]	0.05 [97.4%]
	(0.01)	(0.16)

Main specification

Synthetic controls through January to May



◂ Temporal displacement

Heterogeneity

→ Results are driven by fewer fires during the day; no effect at night

→ Effects seem to be linked to large forest fires

- No effect in croplands, large effect in forested pixels
- Largest effects in provinces with large forest areas, and in areas with low agricultural production
- Effects are larger in magnitude for high intensity fires and fires far from urban areas and roads

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Methodology (1)

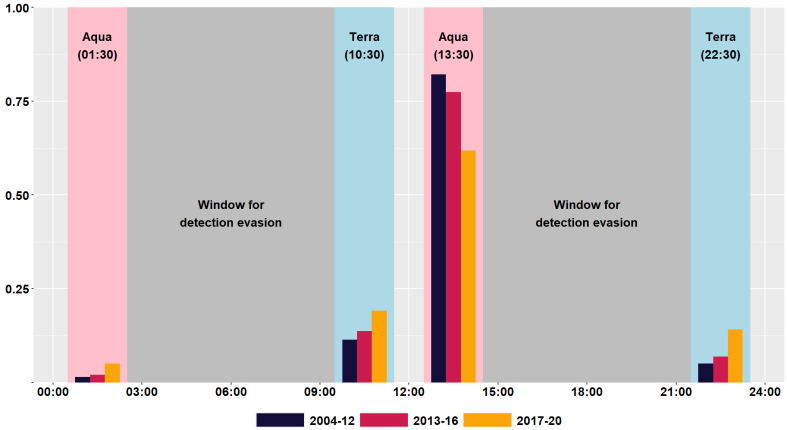
→ FIRMS is my outcome variable and is used for monitoring, so may ignore evasion

- Qualitative evidence that farmers shifted burning times to avoid satellite detection [Mostafanezhad and Evrard (2021)]

→ Evasion should minimize chances of detection

- Longest period between satellites overpasses

Timing of satellite overpasses



Methodology (2)

→ I assume that in the absence of detection avoidance, the share of fires spotted in each satellite overpass should remain constant

→ Counterfactual reduction - multiply pre-treatment share with the total reduction

- I focus on Aqua's daytime overpass

→ Divide the gap between the counterfactual and the true reduction by the total reduction

→ A lower bound of the estimated share of reduction driven by evasion

Estimation

Period	s^D_{Aqua} (1)	$\Delta F_{reduced}$ (2)	$\widehat{\Delta F^D_{Aqua}}$ (3)	ΔF^D_{Aqua} (4)	ρ (5)	γ^D_{Aqua} (6)
Jan-May	0.813	-3.803	-3.091	-3.497	-0.406	0.106
January	0.797	-0.049	-0.039	-0.026	0.013	0.000
February	0.795	-0.569	-0.452	-0.488	-0.036	0.063
March	0.835	-2.394	-1.999	-2.179	-0.180	0.075
April	0.852	-0.303	-0.258	-0.303	-0.045	0.148
May	0.735	0.000	0.000	0.185	0.185	0.000

At least a tenth of the estimated reduction (28%) captures detection evasion

Conclusion

→ I find a reduction in detected fires when seasonal bans are enforced

→ I find temporal displacement

- Increase in fires when the bans are over
- Evidence of changing of timing of fires within the day and evasion from satellites

→ Room for research on welfare effects

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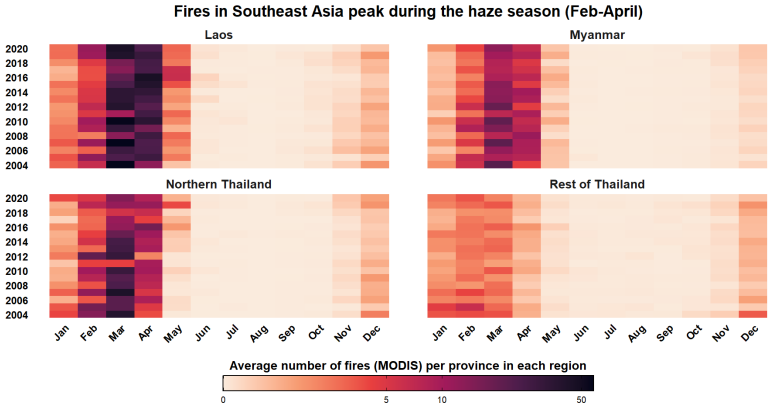
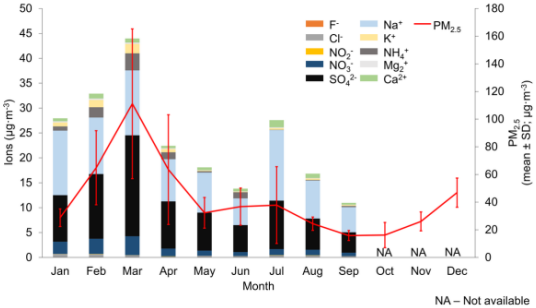
Fires by Month

In each region



Pollution

In Chiang Mai

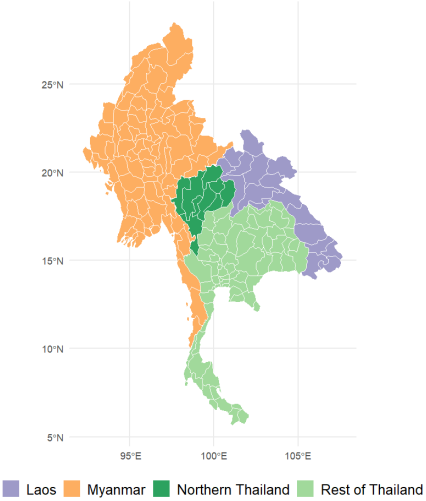


[Nikam, Archer, and Nopsert (2021) and Ponsawansong et al. (2023)]

Ban Dates

Province	2018		2019		2020	
	Start	End	Start	End	Start	End
Chiang Mai	01.03	20.04	01.03	30.04	10.01	30.04
Phrae	15.02	15.04	15.02	15.04	01.02	30.04
Nan	15.02	15.04	15.02	15.04	15.02	15.04
Phayao	16.02	16.04	15.02	15.04	01.03	30.04
Tak	10.02	10.04	01.03	30.04	01.03	30.04
Lamphun	20.02	20.04	01.03	30.04	01.03	30.04
Lampang	10.02	10.04	10.02	10.04	01.03	30.04
Mae Hong Son	01.03	30.04	01.03	30.04	15.03	30.04
Chiang Rai	12.02	12.04	10.02	15.04	-	-

Area of interest



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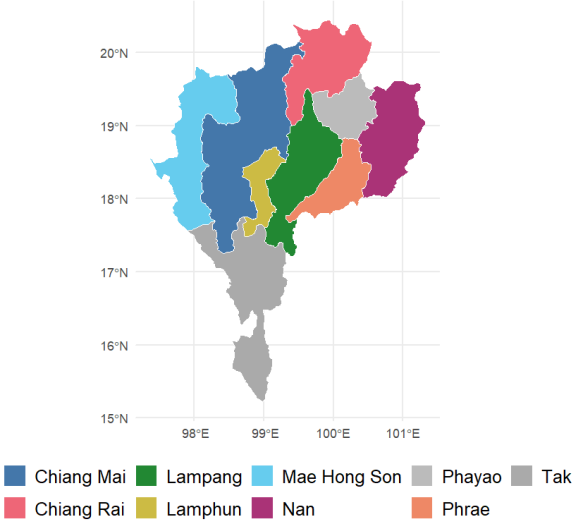
Treated provinces



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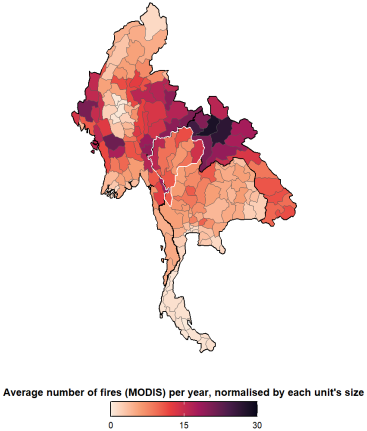
Fires by provinces



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Descriptive statistics

	Country / Region			
Category	Laos	Myanmar	North Thailand	Rest of Thailand
*Yearly Fire Activity (MODIS)**				
Total Fires	16.5 (6.2)	8.93 (6.1)	10.1 (3.2)	3.90 (2.8)
Day Fires	15.7 (6.0)	8.21 (5.6)	9.15 (2.9)	3.45 (2.5)
Night Fires	0.81 (0.52)	0.72 (0.72)	0.95 (0.41)	0.45 (0.37)
Crop Fires	0.20 (0.30)	0.64 (0.96)	0.47 (0.47)	2.29 (2.4)
Forest Fires	15.6 (6.6)	7.18 (5.9)	8.73 (3.5)	0.86 (1.3)
Regional Characteristics				
Area (100 km^2)	135.0 (12.2)	83.8 (6.4)	117.0 (19.3)	60.4 (5.2)
Crop Share	0.05 (0.016)	0.22 (0.024)	0.10 (0.024)	0.34 (0.031)
Forest Share	0.87 (0.031)	0.58 (0.036)	0.76 (0.037)	0.34 (0.032)

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Finding the weights

Solving the following optimization problem:

$$\min_{W_i} \|X_i - \sum_{j=1}^{N_c} w_{ij} X_j\|_V^2 + \lambda \sum_{j=1}^{N_c} w_{ij} \|X_i - X_j\|_V^2$$

Subject to:

$$w_{ij} \geq 0, \quad \text{and} \quad \sum_{j=1}^{N_c} w_{ij} = 1$$

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Estimating ATE

The counterfactual is:

$$\hat{Y}^0_{i,t} = \sum_{j=1}^{N_c} w_{ij}(\lambda) Y_{j,t}$$

The ATE at time t is:

$$\hat{\tau}_t(\lambda) = \frac{1}{N_{treated}} \sum_{i=1}^{N_{treated}} \left(Y^1_{i,t} - \hat{Y}^0_{i,t} \right)$$

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Test statistic

The ratio of mean squared prediction errors after the treatment (separately in each enforcement period), compared to the MSPEs before the treatment

$$\hat{T} = \frac{MSPE_{\text{post}}}{MSPE_{\text{pre}}} = \frac{\frac{1}{|\mathcal{T}_1| N_{treated}} \sum_{t \in \mathcal{T}_1} \sum_{i=1}^{N_{treated}} (Y_{it} - \hat{Y}_{i,t}^0)^2}{\frac{1}{|\mathcal{T}_0| N_{treated}} \sum_{t \in \mathcal{T}_0} \sum_{i=1}^{N_{treated}} (Y_{it} - \hat{Y}_{i,t}^0)^2}$$

Evaluating the effect by treatment status

Estimation

→ I estimate treatment effects separately for provinces currently each treatment status k

Compute a weighted average of the period-specific estimates τ_k, weights correspond to the number of treated units-half months

$$\hat{\tau}_k = \frac{\sum_{h,y} n_{k,h,y} \cdot \hat{\tau}_{k,h,y}}{\sum_{h,y} n_{k,h,y}}$$

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Evaluating the effect by treatment status

Results

Ban Status	Period	
	Stricter enforcement (2017-2020)	Weak enforcement (2015-16)
During	-6.44	-1.68
	(0.16)	(0.26)
	[n=161]	[n=67]
Just after	1.77^{*}	1.46
	(0.08)	(0.14)
	[n=35]	[n=14]
Just before	-0.77	-0.18
	(0.28)	(0.43)
	[n=34]	[n=14]

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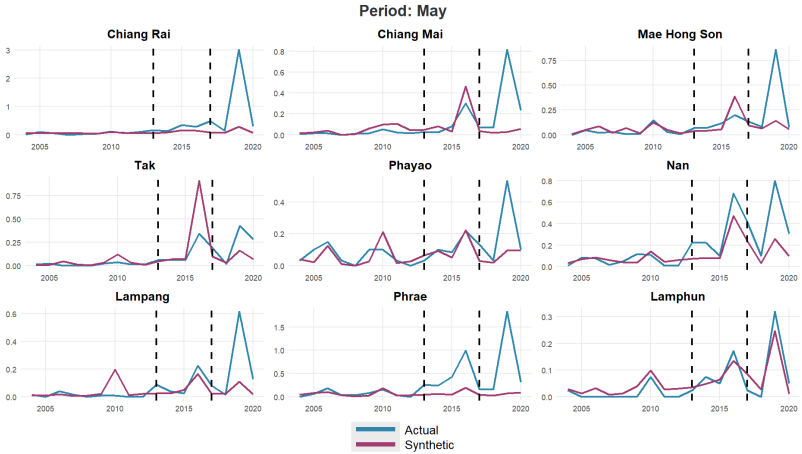
Temporal displacement

Synthetic control plots for May



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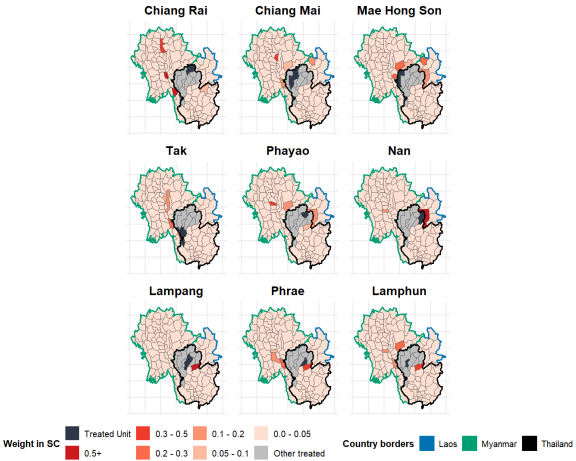
Weights



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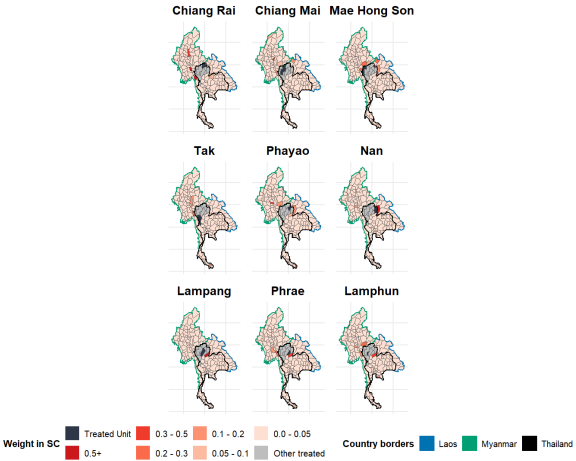
Full map of weights

Weights



▸ Smaller map of weights

Robustness tests (1)

Period	Robustness Tests				
	Baseline	Pre-2008-2012	No Repeated	High Confidence	No Myanmar
	(1)	(2)	(3)	(4)	(5)
Jan–May	-3.24^{*}	-3.51^{*}	-3.10^{**}	-1.85^{*}	-3.17^{*}
	(0.08)	(0.07)	(0.04)	(0.07)	(0.08)
January	-0.07	-0.02	-0.07	-0.04	-0.09
	(0.45)	(0.39)	(0.45)	(0.33)	(0.29)
February	-0.62	-0.53	-0.56	-0.23	-0.27
	(0.19)	(0.26)	(0.14)	(0.33)	(0.70)
March	-2.42^{*}	-2.39^{*}	-2.05^{*}	-1.46^{*}	-1.49
	(0.08)	(0.06)	(0.06)	(0.07)	(0.16)
April	0.18	-0.03	0.09	-0.10	0.05
	(0.14)	(0.16)	(0.10)	(0.13)	(0.15)
May	0.29^{**}	0.26^{**}	0.24^{***}	0.09^{**}	0.19^{**}
	(0.01)	(0.01)	(0.01)	(0.03)	(0.01)

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Robustness tests (2)

Period	Robustness Tests			
	No Laos	No Thailand	No Climate	Fixed Weights
	(6)	(7)	(8)	(9)
Jan–May	-3.21^{**}	-3.37^{*}	-3.00^{**}	-3.24^{**}
	(0.03)	(0.06)	(0.04)	(0.04)
January	0.00	-0.11	-0.05	0.02
	(0.44)	(0.43)	(0.52)	(0.59)
February	-0.59	-0.55	-1.08	-0.19
	(0.12)	(0.15)	(0.18)	(0.77)
March	-2.75^{**}	-2.89^{*}	-2.23^{*}	-1.74
	(0.03)	(0.07)	(0.07)	(0.25)
April	0.08	-0.28	0.08	-1.50
	(0.12)	(0.19)	(0.16)	(0.37)
May	0.28^{**}	0.27^{**}	0.30^{**}	0.17^{*}
	(0.01)	(0.02)	(0.01)	(0.05)

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Robustness tests (3)

Backdating tests

Period	Backdate to 2009 (1)	Backdate to 2010 (2)	Backdate to 2011 (3)
Validation	-0.55	0.46	1.58
	(0.46)	(0.55)	(0.36)
Weak Enforcement	-1.34	-1.03	-0.99
	(0.28)	(0.29)	(0.27)
Strong Enforcement	-3.36^{*}	-3.10^{*}	-3.07^{*}
	(0.10)	(0.08)	(0.07)

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Heterogeneity - day and night

Period	Total fires	Satellite		Time	
		Aqua	Terra	Day	Night
	(1)	(2)	(3)	(4)	(5)
Jan–May	-3.24^{*} [-28.0%]	-3.32^{**} [-40.6%]	-0.16 [-0.5%]	-3.75^{**} [-34.1%]	0.01 [24.9%]
	(0.08)	(0.04)	(0.18)	(0.02)	(0.05)
January	-0.07 [37.4%]	-0.02 [23.1%]	-0.03 [27.9%]	-0.05 [-1.4%]	-0.00 [304.1%]
	(0.45)	(0.49)	(0.34)	(0.49)	(0.24)
February	-0.62 [-17.4%]	-0.53 [-22.4%]	-0.11 [-13.2%]	-0.54 [-21.9%]	-0.05 [340.6%]
	(0.19)	(0.22)	(0.19)	(0.22)	(0.19)
March	-2.42^{*} [-20.6%]	-2.14^{*} [-23.7%]	-0.29 [-2.0%]	-2.39^{*} [-24.2%]	0.03 [172.2%]
	(0.08)	(0.07)	(0.16)	(0.06)	(0.13)
April	0.18 [88.0%]	-0.22 [6.6%]	0.36^{*} [44936.2%]	-0.27 [2.5%]	0.16^{*} [192.3%]
	(0.14)	(0.14)	(0.08)	(0.13)	(0.07)
May	0.29^{**} [373.8%]	0.19^{**} [381.6%]	0.07^{**} [432.7%]	0.24^{**} [235.2%]	0.01 [924.1%]
	(0.01)	(0.01)	(0.01)	(0.01)	(0.17)

Heterogeneity - land Cover

Depending on the classification of the majority of pixels within 1 km, Potapov et al. (2022)

2*Period	Forest		Cropland	
	Strict (2017-2020)	Weak (2013-2016)	Strict (2017-2020)	Weak (2013-2016)
	(1)	(2)	(3)	(4)
Jan–May	-2.81^{**}	-0.80	-0.23	-0.16
	(0.05)	(0.27)	(0.79)	(0.77)
January	0.08	0.07	-0.04	-0.06
	(0.32)	(0.72)	(0.89)	(0.80)
February	-0.04	0.14	-0.07	-0.04
	(0.52)	(0.61)	(0.83)	(0.87)
March	-1.50	0.39	-0.11	-0.04
	(0.16)	(0.40)	(0.28)	(0.42)
April	-1.49	-1.37	-0.02	-0.02
	(0.22)	(0.52)	(0.37)	(0.57)
May	0.14^{**}	-0.02	0.00	-0.00
	(0.03)	(0.18)	(0.33)	(0.77)

Heterogeneity - distance from Roads

Measured as distance from roads based on OpenStreetMap contributors (2025)

2*Period	Province		Country	
	Below median	Above median	Below median	Above median
	(1)	(2)	(3)	(4)
Jan–May	-1.74^{*}	-1.85^{*}	-1.44^{**}	-1.99^{*}
	(0.05)	(0.05)	(0.03)	(0.09)
January	-0.04	-0.01	-0.02	-0.06
	(0.36)	(0.43)	(0.49)	(0.37)
February	-0.40	-0.31	-0.34^{*}	-0.48
	(0.21)	(0.10)	(0.08)	(0.18)
March	-1.08^{*}	-1.29^{*}	-0.63	-1.34
	(0.07)	(0.05)	(0.21)	(0.11)
April	-0.05	0.11^{*}	0.01	0.06
	(0.14)	(0.08)	(0.11)	(0.15)
May	0.16^{**}	0.12^{**}	0.13^{**}	0.15^{**}
	(0.01)	(0.02)	(0.02)	(0.03)

Heterogeneity - distance from Urban Centres

Measured using distance to semi-urban or urban, based on Schiavina, Melchiorri, and Pesaresi (2023)

2*Period	Province		Country	
	Below median	Above median	Below median	Above median
	(1)	(2)	(3)	(4)
Jan–May	-1.50^{*}	-2.03^{**}	-1.32^{**}	-2.78^{*}
	(0.07)	(0.04)	(0.05)	(0.06)
January	-0.06	-0.01	-0.07	0.04
	(0.27)	(0.63)	(0.32)	(0.42)
February	-0.20	-0.33	-0.37	-0.28
	(0.32)	(0.14)	(0.19)	(0.10)
March	-0.89	-1.56^{**}	-0.48	-1.95^{*}
	(0.13)	(0.05)	(0.16)	(0.08)
April	0.04^{*}	0.12	0.09^{**}	0.03
	(0.06)	(0.21)	(0.02)	(0.17)
May	0.12^{**}	0.12^{**}	0.13^{***}	0.13^{**}
	(0.02)	(0.04)	(0.01)	(0.05)

Heterogeneity - by Province (1)

Period	Chiang Mai (1)	Chiang Rai (2)	Lamphun (3)	Lampang (4)	Mae Hong Son (5)
Jan-May	-2.12 [-29.1%]	-1.55^{*} [-22.4%]	-1.52 [-15.0%]	0.56 [21.1%]	-6.18^{*} [-37.2%]
	(0.27)	(0.01)	(0.36)	(0.52)	(0.08)
January	0.08 [36.3%]	-0.23^{*} [8.5%]	-0.34 [-46.0%]	0.11 [9.6%]	-0.00 [411.0%]
	(0.55)	(0.07)	(0.74)	(0.70)	(0.48)
February	-0.16 [11.3%]	-1.94 [-76.3%]	-1.07^{*} [-15.4%]	-0.42^{*} [-7.5%]	0.09 [9.3%]
	(0.26)	(0.12)	(0.10)	(0.09)	(0.95)
March	-1.92 [-44.3%]	-3.54^{**} [-75.7%]	1.24 [100.0%]	0.99 [69.6%]	-4.81^{**} [-46.2%]
	(0.37)	(0.03)	(0.21)	(0.14)	(0.05)
April	-0.14 [-8.7%]	1.45^{***} [80.0%]	-0.14 [-19.5%]	0.67 [592.0%]	-0.43 [-11.8%]
	(0.41)	(0.01)	(0.82)	(0.14)	(0.90)
May	0.26^{**} [928.0%]	0.86^{***} [481.0%]	0.01 [48.9%]	0.17 [334.0%]	0.20^{**} [149.0%]
	(0.03)	(0.01)	(0.36)	(0.15)	(0.02)

Heterogeneity - by Province (2)

Period	Nan (6)	Phayao (7)	Phrae (8)	Tak (9)
Jan-May	-7.42 [-51.8%]	-3.44^{*} [-50.9%]	-1.98 [-25.2%]	-5.53^{***} [-41.4%]
	(0.14)	(0.06)	(0.13)	(0.01)
January	-0.09 [-32.6%]	-0.38 [-64.2%]	-0.07 [-36.0%]	0.28 [49.9%]
	(0.40)	(0.20)	(0.75)	(0.41)
February	-0.55 [-25.7%]	-0.83 [-70.3%]	-0.41 [-15.1%]	-0.25 [33.3%]
	(0.31)	(0.22)	(0.20)	(0.20)
March	-6.65 [-70.3%]	-1.45 [-46.1%]	-0.26 [-3.3%]	-5.39^{***} [-69.4%]
	(0.10)	(0.18)	(0.59)	(0.01)
April	-0.24 [-4.8%]	0.56 [110.0%]	0.58 [51.1%]	-0.73^{**} [4.0%]
	(0.28)	(0.26)	(0.12)	(0.05)
May	0.25^{*} [182.0%]	0.14 [218.0%]	0.56^{***} [895.0%]	0.14 [129.0%]
	(0.07)	(0.14)	(0.01)	(0.12)

Heterogeneity - by satellite's overpass

Aqua at 1:30, Terra at 10:30 (AM and PM)

Period	Aqua		Terra	
	Night	Day	Night	Day
	(3)	(4)	(6)	(7)
Jan–May	-0.05^{**}	-3.50^{**}	0.06	-0.26
	(0.03)	(0.02)	(0.13)	(0.13)
January	0.00	-0.03	-0.00	-0.02
	(0.28)	(0.52)	(0.20)	(0.36)
February	-0.03	-0.49	-0.06	-0.05
	(0.17)	(0.22)	(0.25)	(0.23)
March	0.02	-2.18^{*}	0.03	-0.21
	(0.10)	(0.06)	(0.16)	(0.11)
April	0.07^{*}	-0.30	0.24^{*}	0.03
	(0.08)	(0.15)	(0.05)	(0.20)
May	0.00	0.19^{**}	0.02	0.05^{**}
	(0.18)	(0.01)	(0.16)	(0.04)

Heterogeneity - smaller fires

Measured using VIIRS fires with the MODIS weights

Period	Strict (2017-2020) (1)	Weak (2013-16) (2)
Jan-May	−14.565	−5.921
	(0.175)	(0.187)
January	0.635	0.400
	(0.235)	(0.349)
February	−4.678	−2.024
	(0.199)	(0.205)
March	−2.172	5.424
	(0.337)	(0.380)
April	2.402	6.679
	(0.404)	(0.386)
May	2.110	0.968
	(0.193)	(0.253)

Heterogeneity - crop production

Measured based on production in International Food Policy Research Institute (2019), Q1 is the lowest

Period	Q1	Q2	Q3	Q4
	(1)	(2)	(3)	(4)
Jan-May	-1.96^{*}	-1.49	-0.39	-0.11
	(0.07)	(0.11)	(0.21)	(0.11)
January	0.04	-0.04	-0.05	-0.03
	(0.25)	(0.42)	(0.24)	(0.58)
February	-0.07	-0.15	-0.12	-0.08
	(0.30)	(0.36)	(0.13)	(0.19)
March	-1.50^{*}	-0.96	-0.08	-0.04
	(0.08)	(0.14)	(0.24)	(0.23)
April	-0.14	0.02^{*}	0.10^{*}	0.01
	(0.24)	(0.09)	(0.06)	(0.24)
May	0.08^{*}	0.11^{**}	0.06^{*}	0.03^{**}
	(0.07)	(0.02)	(0.07)	(0.03)

Heterogeneity - dominant crop

Measured based on the crop with highest production in International Food Policy Research Institute (2019)

Period	Maize (1)	Rice (2)	Cassava (3)
Jan-May	-0.029^{**}	-1.375^{**}	-0.131
	(0.018)	(0.048)	(0.133)
January	-0.003^{*}	-0.090	0.004
	(0.066)	(0.169)	(0.175)
February	-0.005^{*}	-0.425	-0.011
	(0.072)	(0.139)	(0.331)
March	-0.013^{**}	-0.667^{*}	-0.140^{**}
	(0.048)	(0.096)	(0.018)
April	-0.002	0.057	-0.006
	(0.807)	(0.151)	(0.193)
May	0.000^{*}	0.105^{**}	-0.007
	(0.054)	(0.030)	(0.133)

Heterogeneity - intensity of fires

Measured using the FRP, below or above median

Period	Low intensity (1)	High intensity (2)
Jan-May	-1.20^{*}	-2.51^{*}
	(0.05)	(0.07)
January	-0.05	-0.03
	(0.43)	(0.49)
February	-0.34	-0.27
	(0.11)	(0.27)
March	-0.65^{*}	-1.90^{*}
	(0.08)	(0.08)
April	0.24	-0.19
	(0.10)	(0.13)
May	0.18^{**}	0.10^{***}
	(0.02)	(0.01)